

A Citation-Calibrated Model of the Downstream Consequences of Worker Displacement

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Parameter set v1.48

Abstract

We present `downstream`, an open-source model that propagates worker-displacement exposure through a ledger of published effects. It distinguishes earnings gaps, direct effects, relative risks, and jobs-per-job multipliers. Each parameter retains its citation, evidence precision tier, population scope, and uncertainty band. Scenario counts require verified baselines. Parameter sampling, exact moment calculations, sensitivity analysis, and named structural variants expose uncertainty in both estimates and composition choices. Current China-shock retrodictions show mixed results: family-status contrasts are covered, while male mortality variants overpredict and lower-quartile earnings losses remain underpredicted after a wage-spillover channel is added. Independent event back-tests remain blocked on data, and prospective forecasts have not yet been registered. We report those limits alongside the implemented methods. Outputs are conditional population-average estimates, not predictions about individuals or proof that an exposure was caused by a particular filing or employer. A current mathematical review identifies mortality transport, baseline, and place-counterfactual limits; numerical consistency alone does not establish causal validity.

1 Introduction

Worker displacement has consequences beyond a worker’s immediate earnings. The underlying studies measure different outcomes in different populations: worker earnings, mortality, and children’s later earnings are prominent examples [41, 64, 52]. A combined model must state how those estimates can be used together. A citation alone does not justify multiplication: a jobs-per-job multiplier and an intergenerational earnings elasticity act on different quantities.

This model makes those distinctions explicit. It provides a typed ledger, a versioned parameter set, reproducible uncertainty calculations, and validation results that retain misses. Its present implementation is a conditional propagation model with population aggregation. It does not simulate individual labor-market matching, migration, or behavioral interactions. The term microsimulation describes the intended family-level use; it should not imply that the code currently advances a representative synthetic population through an annual life cycle.

2 Relation to other modeling approaches

2.1 Life-course and household microsimulation

The intended person-and-place simulator belongs to an established research tradition. SimPaths projects individual and household histories across work, family relationships, health, and finances in an open-source framework [14]. It is a close architectural reference for dynamic feedback and household context. LifeSim combines developmental, economic, social, and health outcomes across an English birth cohort’s lifespan [60]. It is especially relevant to linking childhood circumstances with later outcomes. Neither project should be reduced to a collection of isolated outcome multipliers.

Urban Institute’s DYNASIM ages individuals and families through demographic and economic transitions, including marriage, divorce, work, earnings, disability, and retirement [65]. The Future Elderly Model connects health and economic outcomes to evaluate alternative health and policy scenarios [66]. These programs provide useful precedents for baseline trajectories, family structure, scenario analysis, and empirical evaluation. Their domains and target populations differ; this comparison is selective, not a systematic ranking of model quality.

Downstream currently has a narrower implementation: cited response coefficients and count conversions conditional on an externally supplied exposure. Its proposed contribution is an inspectable connection between evidence applicability, composition restrictions, uncertainty, and accessible scenario comparison. Openness, simulation of multiple domains, and citation of evidence alone do not establish novelty. Whether its composition rules improve out-of-sample performance remains a research question, not a demonstrated advantage over these predecessors.

2.2 Causal composition and population consequences

Our design separates three tasks often combined in larger models: estimating an exposure, propagating its consequences, and evaluating the result. The exposure is supplied by the caller. A dynamic population model would additionally need entry, exit, and transition processes that this implementation does not estimate. An interacting-agent model would need behavioral rules and a network; neither is present here.

Regional multipliers enter as a separate count boundary, following the local-employment literature [51]. They are not passed through an earnings chain. Evidence synthesis enters through published individual studies and meta-analyses [54, 31]. We compose selected links under declared assumptions rather than pool unlike estimands into a single effect. The model is therefore an auditable synthesis of conditional responses, not a newly identified causal estimate of the entire system.

3 Model and evidence

Figure 1 separates supplied inputs, model operations, and reported outputs. Its arrows describe computation; they do not establish that a causal pathway is identified.

3.1 Exposure and outcomes

A scenario supplies displaced-worker exposure N , children per exposed worker C , a tradable-job share s , and a mortality window Y . The worker module reports earnings and mortality responses. The child module reports earnings across three descendant generations. Family and community modules expose additional pathways, but their availability in a module does not imply that the scenario count interface reports them. The current count interface returns local service jobs, excess deaths, and child lifetime earnings losses, subject to the required baselines. The mortality count is a reference scenario for US men aged 45–54. Its displacement effect comes from a study of high-seniority men. The child earnings link comes from Canadian father–son firm closures and uses a US male lifetime-earnings baseline in 2024 dollars. Applying either result to a mixed or unknown population requires a separate transport argument. New receiving-community crime links in this parameter set start from an immigrant-influx input. They are a separate evidence family, not a consequence inferred from displaced-worker exposure.

The entity adapter accepts either a documented exposure or a warehouse aggregate with a stored total and components. The warehouse mapping checks agreement between the headline and breakdown totals. It checks that the worker-equivalent base, wage-depression, and applicable WARN components reconcile within the upstream rounding tolerance. It preserves the total; it does not re-estimate a wage-gap formula or add a notice count to a worker count. Source and method references accompany the result. Person subjects use the same population-average impact framing as employers. This interface does not establish a causal link between filings and worker displacement.

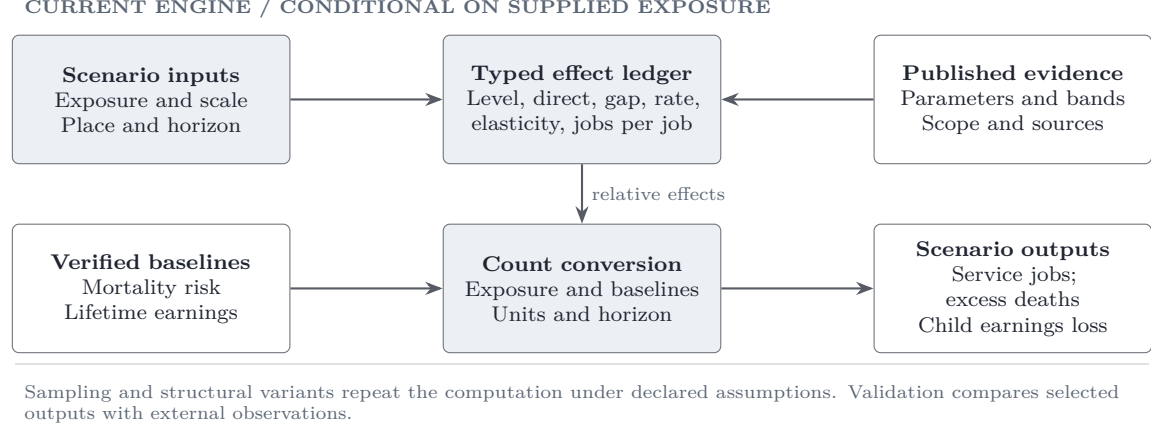


Figure 1: The current calculation boundary. Exposure is supplied, not inferred. The ledger separates effect types before count conversion. A citation records provenance; it does not by itself establish transportability or causal composition.

3.2 Typed composition

Let v be a counterfactual-relative earnings level, with no loss represented by $v = 1$. Let p be a parameter and t an intergenerational transmission coefficient. The ledger distinguishes

$$\text{level: } v' = vp, \quad (1)$$

$$\text{direct: } v' = p, \quad (2)$$

$$\text{gap: } v' = 1 - t(1 - v). \quad (3)$$

A rate step records a relative effect for later conversion against a baseline. A level ratio multiplies a count outside the chain. An elasticity converts an input percentage change into an outcome percentage change. These distinctions prevent a transmission coefficient from being mistaken for a relative earnings level.

For the current parameter points, the direct child level is 0.9076, the grandchild level is 0.9492, and the great-grandchild level is 0.972. These values come from the same code used by the CLI; the first is the direct child estimate and the later values propagate its gap [52, 62, 26, 21, 45, 1]. The father’s earnings shock is not multiplied into the directly estimated child effect. Repeated transmission beyond the identified generations remains a modeling assumption.

Even the first transmission requires an extra causal assumption: a between-family income association need not equal the response to a displacement intervention. If t is instead interpreted literally as a constant log-log elasticity, the finite-change mapping is $v' = v^t$. The implemented gap rule is its first-order expansion about $v = 1$, $v^t = 1 + t(v - 1) + O((v - 1)^2)$. Thus it is an explicit linear approximation, not an exact conversion of a log elasticity. A future structural variant should compare these mappings without treating either as independently identified transmission of a displacement effect.

Parallel divorce and family-size pathways are not silently added to the displacement pathway. The same affected children may appear in several studies, and summation can double-count their losses. A named structural variant exposes gap-additive composition as an experiment rather than making it the default.

3.3 Parameter custody

This build contains 67 parameter rows in set v1.48. Appendix B is generated directly from the parameter file. Precision tiers describe how the number was extracted: **EXACT** denotes a pinned table or full-text estimate; **abstract/results** variants identify the narrower source read; **canonical** marks an established value with remaining extraction work. Precision of extraction is distinct from strength of causal identification. Associational conversion factors and parallel evidence must retain their limitations even when transcribed exactly.

Each row also stores population scope and notes on its band. Bands can represent reported uncertainty, a range across specifications or studies, or an explicitly declared approximation. They are not interchangeable confidence intervals. The audit checks citation resolution, units, and parameter structure. It cannot prove that a source has been interpreted correctly. Full notes and evidence classes remain in the CSV and bibliography rather than being replaced by the compact appendix.

3.4 Count conversion

With annual mortality baseline m , source-offset odds ratios r_j , and lifetime earnings baseline L , the current scenario implements

$$q(r) = \frac{rm}{1 - m + rm}, \quad (4)$$

$$D = N \left\{ (1 - m)^Y - \prod_{j \in \mathcal{P}} [1 - q(r_j)]^{d_j} \right\}, \quad (5)$$

$$E = NC(1 - g_c)L, \quad (6)$$

$$J = Ns\ell, \quad (7)$$

where $\mathcal{P} = \{0, 1, 2:3, 4:5, 6+\}$ indexes the displacement-year and source-offset phases, and their durations are $d_0 = \min(Y, 1)$, $d_1 = \min\{\max(Y - 1, 0), 1\}$, $d_{2:3} = \min\{\max(Y - 2, 0), 2\}$, $d_{4:5} = \min\{\max(Y - 4, 0), 2\}$, and $d_{6+} = \max(Y - 6, 0)$. Here g_c is the child earnings level and ℓ is the service-job multiplier. Engine 0.2 converts the stored odds ratios to risks and accounts for survival depletion. The follow-up includes the displacement year as year 1. The default **source_profile** uses all five source-offset phases. The source working paper’s Table 5¹ reports the displacement year, offsets +1, +2–3, and +4–5, and +6 and later. Under this convention, the +6 coefficient begins in follow-up year 7. The historical **source_aligned** option preserves the former incomplete three-phase approximation for reproducibility only. The optional **immediate_sustained** timing applies the +6 coefficient after the displacement year as a sensitivity assumption. The historical additive formula remains available as **legacy_additive**. Current V1 and scoreable V2 mortality calculations use the production survival kernel and the source-offset profile by default.

Only verified baseline values enter a count. A missing baseline produces a named blocked outcome, or raises an exception in strict mode. The baseline file records source, population, units, and derivation. Table 1 lists its current values; some support modules beyond the count interface.

Table 1: Stored baselines. Units and source qualifications are discussed below.

Baseline	Value	Population
all_cause_mortality_annual	0.004944	US prime-age men (45-54)
divorce_5y_cumulative	0.1045	US married couples 5y window
ipv_annual_incidence	0.0027	US residents age 12 or older
youth_crime_participation	0.055948	US young men 18-24
median_male_lifetime_earnings	2,591,418	US male median lifetime earnings

Mortality uses CDC WONDER deaths per person-year as an approximation to the annual probability m . A person-time rate is not itself a cohort probability [18]; under a constant hazard λ , that probability is $1 - e^{-\lambda}$. The current implementation applies this conversion for a compatible mortality rate, but does not advance age-specific baselines. Divorce uses the Census SIPP five-year marriage-survival complement, which is not a pure divorce incidence measure. The NCVS IPV rate counts incidents rather than unique victims. The youth-crime baseline counts arrests per young man rather than unique offenders. Lifetime earnings are in 2024 dollars, derived from SSA age-band earnings and BLS inflation data; summing cross-sectional medians approximates, rather than equals, the median lifetime total. Exact source tables, windows, and arithmetic are retained in **params/baselines.csv**.

¹Sullivan and von Wachter, NBER Working Paper 13626, PDF page 54: https://www.nber.org/system/files/working_papers/w13626/w13626.pdf.

3.5 Place resolution

Where measured place values exist in compatible units, baseline pooling uses $b_* = (1 - w)b_{\text{national}} + wb_{\text{local}}$ with $w = n/(n + k)$. The precision count n comes from the input data; the prior precision k is a declared modeling choice. Missing local information keeps the national baseline and reports the reason.

The mobility layer derives a multiplier $M = 1 + 18\gamma(q - q_0)/100$, where q is the county’s permanent-resident income-rank percentile and q_0 is the national reference. The exposure coefficient γ comes from Chetty and Hendren [20]. Applying an income-rank exposure effect as a modifier of a displacement loss is an additional assumption, not the estimand of that study. The count interface uses the same-place contrast $M(1 - g)$, equivalently a child multiplier $1 - M(1 - g)$, so a null displacement effect ($g = 1$) remains null at every place. The child modifier is applied once before transmission. Descendants inherit that adjusted gap. The historical repeated application is available as an explicit structural alternative, not the default. Place-aware sampling recomputes M for each parameter draw, while county measurements and pooling weights remain fixed.

This null-safe formulation does not identify a displacement-by-place interaction. If M changes the earnings baseline in both exposure arms, their level difference would be $M(1 - g)$; if it changes susceptibility to displacement, an independently supported effect modifier is needed. Place-specific child dollar losses remain exploratory and the national child-dollar result is the reference.

4 Uncertainty and experiments

Figure 2 distinguishes the uncertainty objects used below. Each answers a different question.

4.1 Parameter sampling

Latin hypercube sampling stratifies each parameter marginal [49]. Positive ratio units default to log-uniform sampling; other undeclared marginals default to uniform bands. Rows can explicitly select normal or lognormal shapes. The current normal-family sampler clamps draws at the stated band limits. This is a bounded approximation with endpoint mass, not sampling from a renormalized truncated-normal density. The point is retained as the declared center when applicable.

Declared rank correlations can be imposed through an Iman–Conover reordering [39]. The shipped table contains 2 declared pairs linking worker earnings with peak and sustained mortality. Their direction is motivated by the shared recession context, but their moderate negative magnitude is a declared assumption, not an extracted sampling covariance. Monte Carlo applies these pairs by default; analytic chains and standard Sobol experiments retain their independence assumptions. Each Monte Carlo output records the seed, draw count, sampler, and parameter version. Sample quantiles describe uncertainty conditional on the specified distributions; they do not include all model error.

Scenario output separates three objects: endpoint support envelopes, central 90% parameter-only intervals, and simulated realized mortality counts. For an integer cohort, let $S_0 = (1 - m)^Y$ and let S_1 be the exposed survival product above. Conditional on each parameter draw,

$$K_0 \sim \text{Binomial}(N, 1 - S_0), \quad K_1 \sim \text{Binomial}(N, 1 - S_1), \quad K_0 \perp K_1. \quad (8)$$

Then $E[K_1 - K_0 \mid \theta] = N(S_0 - S_1)$. Both cohorts use the same resolved county or national baseline. Their independent difference is a reference-cohort contrast, not a paired individual causal effect. Fractional worker equivalents and the legacy additive mortality method do not receive this count simulation. Although the API calls it “posterior-predictive,” heterogeneous elicited parameter bands are not a fitted joint Bayesian posterior; this is conditional predictive simulation under the declared input distributions.

Support envelope	What do the endpoint calculations span? A bound under the stated calculation; no probability content.
Parameter interval	How do outputs vary across parameter draws? Central 90% under declared distributions; excludes unmodeled error.
Realized mortality counts	How might counts vary between simulated cohorts? Conditional binomial variation; not one person’s paired causal effect.
Structural alternatives	How do selected composition choices change results? Named model variants; no probability distribution over all models.

Figure 2: Four distinct uncertainty summaries. They are not interchangeable intervals. Uncertain exposure, baseline estimation, study transport, and missing mechanisms are not all captured by the current parameter draws.

4.2 Analytic moments and coverage

For an independent level multiplier X , raw moments obey $E[(VX)^k] = E[V^k]E[X^k]$. A direct step replaces the moments. For a gap step with independent T and V ,

$$E[V'] = 1 - E[T] + E[T]E[V], \quad (9)$$

$$E[(V')^2] = 1 - 2E[T] + E[T^2] + 2(E[T] - E[T^2])E[V] + E[T^2]E[V^2]. \quad (10)$$

The implementation also propagates third moments for supported marginals. It uses them for skewness and Cornish–Fisher quantile approximations. Exact moments do not make those approximate quantiles exact. Unsupported distribution cases are refused. The identities also require that each new multiplier be independent of the accumulated quantity; reuse of a parameter can violate that condition.

Monte Carlo comparisons check the analytic calculations. Coverage closure experiments draw synthetic truths under the declared model and measure whether independently built intervals cover them. These tests assess interval machinery under its own assumptions. They cannot establish coverage for real-world events. Correlation stress experiments similarly price a hypothetical dependence choice without treating it as evidence.

4.3 Sensitivity, sweeps, and structural variants

The sensitivity interface estimates first-order and total Sobol indices using a Saltelli-style design [61, 57]. Seed replicates report design variation; that spread is not a confidence interval on the true index. Standard variance attribution assumes independent inputs. A place adjustment can make the mobility coefficient contribute to child-line sensitivity even though it has no effect at the national reference.

A separate correlated-block option groups all connected dependent parameters and swaps each joint block between independently seeded designs. It estimates attribution to independent blocks, not to individual dependent coefficients. Reusing a fixed copula rank template across the two designs invalidates that experiment; the current repair tests it against an additive model with known block variance shares.

A sweep pins one parameter inside its cited band and recomputes scenario counts. A value-of-information experiment narrows each parameter band around its point and compares output-band widths on a common seed. This is a precision-priority heuristic, not an expected utility calculation or a prediction of what a new study will find. Experiments carry a version marker and do not overwrite the published parameter set. Structural ensembles report named alternatives for gap additivity, transmission decay, and closure selection. Their spread represents selected assumptions, not a probability distribution over all possible models.

The policy comparator subtracts caller-supplied exposure scenarios. It reports a conservative difference of support envelopes, with exact cancellation for identical cases. Shared-draw paired contrasts are a future improvement; the current envelope is not a probabilistic policy effect interval and does not identify a policy-to-exposure bridge.

4.4 Evidence synthesis and county likelihoods

The optional synthesis module implements DerSimonian–Laird heterogeneity estimation and inverse-variance random-effects pooling only after exact link, scale, population, and horizon checks, with declared treatment, comparison, outcome, and overlap metadata retained for review. Its intervals use a plug-in normal approximation; uncertainty in the estimated between-study variance is not propagated. Few-study prediction intervals can be unreliable [23]. Matching scale strings alone does not enforce compatible follow-up, population, design, or independent samples. Metadata review remains necessary, and these outputs are not automatically admitted as parameters.

The standalone county module uses $D \mid \lambda \sim \text{Poisson}(T\lambda)$ with a gamma prior, where T is compatible person-time, and reports the constant-hazard conversion $1 - \exp(-\lambda)$ separately from the rate likelihood. It requires national-prior outcome, population, and window metadata to match the county observation. The county module is not integrated into scenario baseline uncertainty.

5 Validation results

5.1 V0: internal consistency

The direct child estimate is 0.9076 [0.844, 0.976], while the worker-shock path composed with intergenerational transmission is 0.89 [0.85, 0.94]. Their bands overlap. This checks compatibility between two modeled paths; it does not provide an independent validation of their causal claims.

5.2 V1: China-shock unit scorecard

Figure 3 plots the unit comparisons. Exact values are retained in Appendix A. The unit scorecard uses the exposure bridge and measured coefficients transcribed from Autor, Dorn, and Hanson [4]. The bridge maps an import shock into manufacturing displacement. The downstream mortality path uses a separate study [64], but comparing it with a commuting-zone male–female mortality differential requires strong population and outcome-transfer assumptions.

The pooled-adult row covers the observed point with its current band. Both male variants overpredict and exclude that point. A measured confidence interval and a modeled parameter band answer different questions; overlap is not proof of validity. The full JSON scorecard also reports whether the modeled point lies inside the measured interval. Age matching does not remove the distinction between an absolute mortality level and a sex differential, nor the transfer from high-seniority displaced workers to a broader exposed population.

The divorce path uses independent displacement studies [56, 19] and a marriage baseline. Its point estimates undershoot the observed family-status stock change at both windows. That stock includes widowhood and separation; divorce incidence alone cannot reproduce all its margins. Remarriage is a candidate missing mechanism, not an extracted explanation. Several additional target outcomes are withheld because the implemented evidence path would reuse the target coefficient rather than provide an independent comparison.

5.3 V1: commuting-zone contrasts and scoring

Figure 4 plots the panel contrasts on separate axes for family status and earnings. The deposited panel extract contains 1444 commuting-zone-period rows. Comparisons demean exposure and outcomes within period before contrasting high- and low-shock terciles. Pooling without that adjustment confounds period differences and changes the sign of the family-status comparison. The tests retain this failure mode.

The family-status contrasts are covered. Direct worker earnings losses alone substantially underpredict the observed lower-quartile earnings change. Adding the non-displaced wage response [6] reduces that gap but does not eliminate it. The earlier direct-only miss remains beside the composed result. The spillover extension was motivated by the observed miss; it is a diagnostic revision, not a fresh held-out success. Shared China-shock evidence further limits independence.

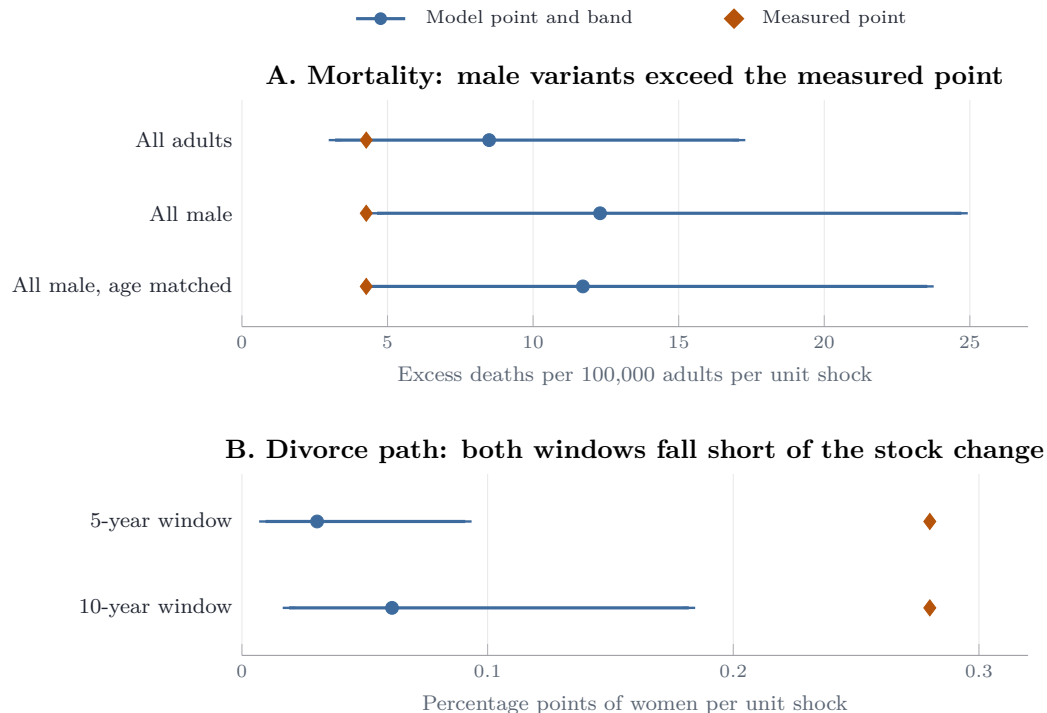


Figure 3: China-shock unit checks generated from the current engine. Blue lines are model bands, not a common confidence level; amber diamonds are measured point targets without their sampling intervals. Panel A compares modeled mortality with a male–female differential; panel B compares divorce with a broader family-status stock change. These target mismatches limit both comparisons. Numerical values appear in Appendix A.

The model median slope is 0.079 against an observed slope of 0.28. CRPS is 0.1779 in percentage-point units; PIT is 1. CRPS summarizes distance between a predictive distribution and an observed point; PIT locates that point within the distribution [36]. Scoring against a published coefficient treats its point estimate as the target and does not integrate its measurement uncertainty. The upper-tail PIT and earnings residual show why selected tercile coverage should not be described as overall model validation.

5.4 V2 and V3: incomplete validation

Independent-event protocols exist for NAFTA, the auto crisis, and BRAC. Their scorecards are blocked on event-specific exposure bridges or measured outcome inputs. A tariff change cannot inherit an import-penetration displacement bridge merely because both concern trade. Descriptive employment loss does not supply a quasi-experimental measured outcome. No completed V2 event score is claimed here.

Prospective registration and scoring checks are implemented, but no real event has yet been registered. It requires forecasts fixed before outcomes are available, a declared exposure and time horizon, and a scoring rule fixed before evaluation. No prospective result is reported. Unit and panel V1 results are the current external checks.

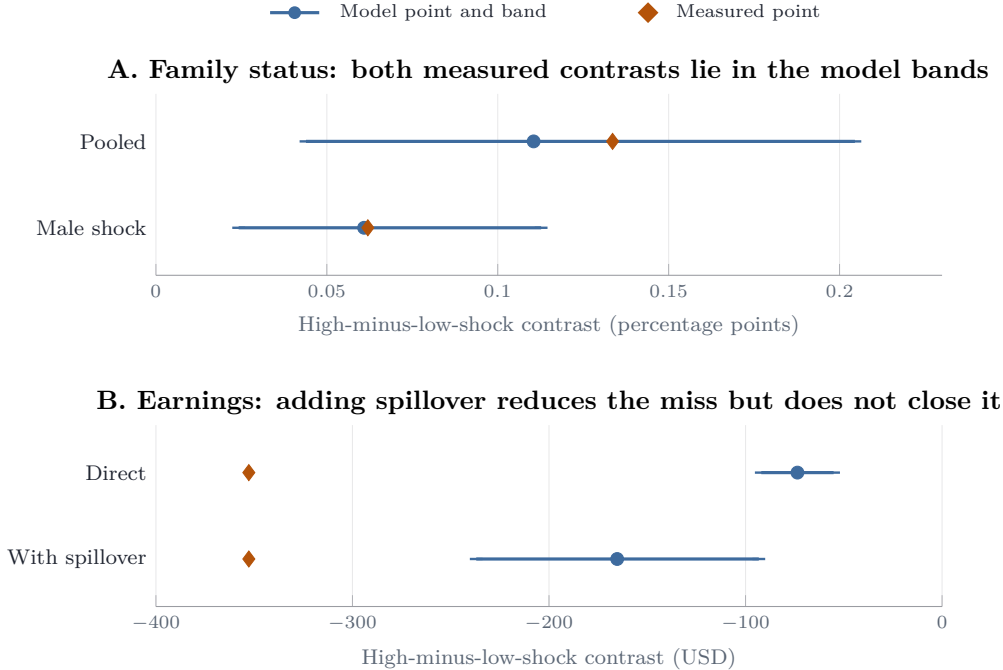


Figure 4: Within-period-demeaned commuting-zone contrasts from 1444 zone-period rows. Model points and bands come from the same generated scorecard as the tables. Measured diamonds omit sampling uncertainty; point coverage alone is not validation. The wage-spillover addition followed the original miss and is not an independent held-out success. Axes retain the outcomes’ different units.

6 Limitations

The model conditions on displacement and does not identify who was displaced or why. An upstream employer exposure can therefore be more uncertain than the downstream parameter band suggests. The entity adapter checks arithmetic and provenance fields, not the factual or causal validity of the supplied records.

Studies differ in country, age, sex, tenure, and measurement window. Transporting them across populations is a substantive assumption. Mortality uses a constant baseline and the five-phase source-offset profile described above. It does not advance age, model individual risk factors, or resolve transport from the source study population. Lifetime earnings conversion uses a synthetic cross-sectional profile. Local employment multipliers omit migration and equilibrium adjustment. These choices can affect magnitudes even when arithmetic and citations are correct.

Transmission across distant generations is weakly identified. Parallel family pathways may overlap, place adjustments transfer a rank effect to an earnings scale, and repeated place steps can magnify the adjustment. None of these is resolved by collecting more Monte Carlo draws.

The current bands use heterogeneous evidence conventions and simplified marginal shapes. Only the declared correlation pairs enter Monte Carlo; other dependencies remain unmodeled. Place measurements and baseline estimates are generally fixed. Structural ensembles cover only selected alternatives. Internal coverage tests therefore validate machinery, while V1 exposes economic and measurement discrepancies. There is no basis for treating the output as a complete uncertainty interval for an individual, employer, or community.

The repository also records refused and contested links. It does not encode a positive immigration-to-crime effect from a pooled literature that does not support one [53, 16]. Observational findings may supply cross-checks without setting a causal point. Named competing specifications must remain visible rather than being averaged into a preferred headline.

7 Prioritized development

The next release should first improve the estimands already used in count headlines. Add uncertainty for the mortality baseline, test transport from the source study population, and assess sensitivity to the constant-hazard conversion. Resolve the place counterfactual before adding further place-driven descendants. Require edge metadata for effect scale, intervention, mediator versus total effect, source population, and time window; compatible units alone do not establish causal composability.

Next, fit likelihoods to actual study estimates and county denominators, with overlap and window checks, explicit baseline uncertainty, and small-sample sensitivity for synthesis. Preserve null, protective, and conflicting effects. Add age/sex/tenure strata only where source estimates support them. For local jobs, separate manufacturing and high-tech multipliers and test the assumption that expansion effects transfer symmetrically to displacement.

Validation should prioritize one completed independent event before expanding the graph: freeze a matched exposure bridge, geography, population, outcome window, and observation model, then report every scored outcome. Register one prospective forecast with an independent timestamp. Numerical tests should include null-intervention invariance, same-place counterfactual cancellation, independently seeded sampling designs, and known analytic variance shares. These checks complement the existing property tests; passing code tests cannot repair an unidentified causal bridge.

7.1 A person-and-place research program

Figure 5 summarizes the proposed comparison and the research gates that precede public use.

The longer-term target is a distribution of possible outcomes conditional on a supplied profile and specified event, with an explicit reference scenario. It requires a baseline model of how employment, income, health, household relationships, and location evolve without that event. Baseline transitions may be estimated from documented longitudinal data; their predictive associations must be distinguished from identified causal intervention effects and from declared structural assumptions.

Personalization should distinguish characteristics that change baseline risks from those supported as effect modifiers. A location field does not establish a county-specific treatment effect. Linked household members, children’s ages, shared resources, and event histories are necessary for family consequences. Community spillovers require an appropriate population model; summing person-level effects does not identify every regional response. Event definitions must include onset, duration, dose, comparison condition, and ordering. Unsupported interactions remain unresolved even when each event has an individual effect estimate.

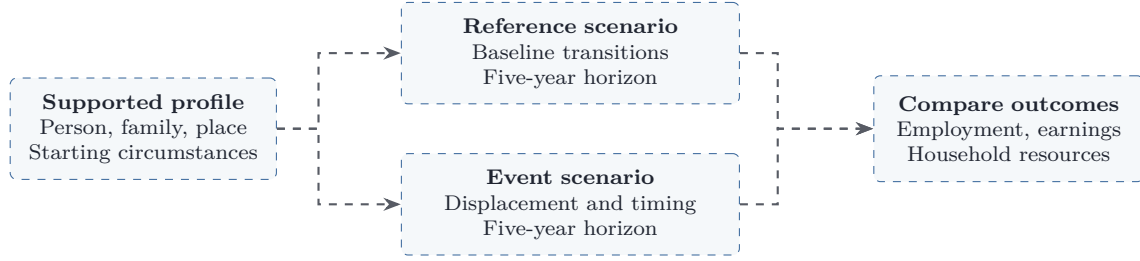
The next substantial demonstration should compare a synthetic household’s five-year employment, earnings, and resource trajectories with and without involuntary displacement, restricted to supported outcomes. A protective intervention can follow when independently justified. Baseline validation, event-effect validation, and comparisons with simple models should precede long multigenerational chains. Shared random draws can improve comparison precision; they do not identify the joint distribution of one person’s potential outcomes. These future capabilities are not claimed for the current engine. Milestones and acceptance criteria are recorded in `docs/LIFE_COURSE_RESEARCH_PLAN.md`.

7.2 Public calculator as a research interface

An eventual website should make the profile, event, comparison, outcome unit, time horizon, and evidence applicability visible together. It should show absolute outcomes and scenario differences, distinguish uncertainty about the model from variation among simulated lives, and separate missing evidence from a supported null effect. Information should unfold from a short comparison to trajectories, assumptions, and study receipts, without requiring the visitor to understand a graph before using the calculator. Neutral scenario colors, legible charts, accessible controls, and clear unsupported-result states are part of scientific communication.

Initial public use should be restricted to evaluated scenarios and supported profiles. A synthetic example can demonstrate interaction before those capabilities are ready, but must be labeled as illustrative. Usability evaluation should measure whether visitors understand the comparison and its limits, rather

PROPOSED / NOT IMPLEMENTED LIFE-COURSE CAPABILITIES



RESEARCH GATES BEFORE PUBLIC USE



Figure 5: The proposed first life-course demonstration and its release gates. Both scenarios start from the same supported profile. The diagram specifies comparisons, not predicted trajectories. Further events, community feedback, and longer horizons require additional evidence and validation.

than treating visual polish or expressed trust as evidence of comprehension. The separate product plan, `docs/WEBSITE_EXPERIENCE_PLAN.md`, specifies the intended journey, visual direction, explanation patterns, and release criteria; it is a design proposal, not a description of a deployed service.

8 Reproducibility and availability

The model and paper source are provided under the repository’s MIT license. Run the tests, `downstream audit`, and `downstream validate` to inspect implementation and evidence checks. Sampling commands record their seeds and draw counts. Parameter exports carry a version; the paper also records a content digest because parameter version alone does not identify engine code changes. The September 10 review repaired county predictive-baseline consistency, disconnected custom chains, ignored chain starting values, and the fixed copula seed used in sensitivity designs. Correlated Monte Carlo results can consequently change at the same historical user seed.

Running `make -C paper` regenerates tables and chart data, builds the PDF, and writes `paper/generated/results.json`. This JSON contains the full validation output and the source SHA-256. The generated appendix preserves parameter points, bands, tiers, and citation keys. Population scope, distribution declarations, and extraction notes remain in the underlying CSV files. Source content digest for this build begins `8d00c8b50a1a1e52`. Generated numerical results are not manually copied into the prose. Vector diagrams and charts use the same restrained palette throughout. Model bands and measured points differ in both color and marker shape. Chart coordinates retain the engine’s unrounded values; the tables provide compact rounded readings.

A Numerical validation scorecards

These tables retain the exact comparison structure behind Figures 3 and 4. Values are rounded to four decimal places; chart coordinates use the unrounded engine output. “Covered” means only that the measured point lies in the model band, not that the model is causally validated.

Table 2: Mortality per 100,000 adults per unit shock. Coverage asks whether the measured point lies in the modeled band.

Variant	Model point [band]	Measured	Covered?
all adults	8.49 [3.2, 17.07]	4.27	yes
all male	12.3 [4.64, 24.71]	4.27	no
all male age matched	11.71 [4.42, 23.54]	4.27	no

Table 3: Unit divorce-path predictions in percentage points of women, compared with the measured family-status stock change.

Window (years)	Model point [band]	Measured
5	0.0306 [0.0096, 0.0909]	0.28
10	0.0611 [0.0192, 0.1819]	0.28

Table 4: Within-period-demeaned panel contrasts. Coverage uses the measured contrast as a point target.

Outcome	Model point [band]	Measured	Covered?
Divorce, pooled (pp)	0.1105 [0.0439, 0.2045]	0.1336	yes
Divorce, male shock (pp)	0.0609 [0.0242, 0.1127]	0.062	yes
Earnings, direct (USD)	-73.59 [-91.99, -55.19]	-352.73	no
Earnings, with spillover (USD)	-165.31 [-237.03, -93.24]	-352.73	no

B Current parameter set

The table is an inventory, not a claim that every row belongs to one composable causal chain. Values use each destination node's declared unit in `params/nodes.csv`. Different rows cannot be compared as if their numbers shared a scale. Parameter values below are rounded to four decimal places; the source CSV retains full precision.

Table 5: Generated parameter inventory, v1.48.

Link	Point [band]	Tier	Sources
displacement->worker_earnings	0.8 [0.75, 0.85]	EXACT-abstract	[41, 52, 29]
earnings_shock->mortality_sustained	1.135 [1.033, 1.247]	EXACT	[64]
earnings_shock->mortality_peak	2.6696 [2.1151, 3.3696]	EXACT	[64]
earnings_shock->mortality_offset_1	1.9838 [1.3672, 2.8777]	EXACT	[64]
earnings_shock->mortality_offsets_2_3	1.7316 [1.3244, 2.2641]	EXACT	[64]
earnings_shock->mortality_offsets_4_5	1.2712 [1.0936, 1.4777]	EXACT	[64]
displacement->divorce_hazard	1.11 [1.05, 1.25]	canonical	[56, 19]
displacement->child_earnings	0.9076 [0.844, 0.976]	EXACT	[52]
divorce->child_earnings	0.95 [0.9, 0.99]	canonical	[34]
family_size->child_earnings	0.98 [0.95, 1]	canonical	[10]
child_earnings->grandchild_earnings	0.55 [0.4, 0.6]	canonical	[62, 26, 21]
grandchild_earnings->greatgrandchild_earnings	0.55 [0.4, 0.6]	canonical	[45, 1]
ipv_exposure->daughter_violence_odds	2.5 [2, 3]	canonical	[67, 30]
displacement->local_service_jobs	5 [1.6, 5]	canonical	[51]
school_spending->child_earnings	1.07 [1.03, 1.1]	canonical	[40]
youth_wages->youth_crime	-1 [-1.5, -0.5]	canonical	[32, 33]
wage_ratio->household_ipv	-0.813 [-1.45, -0.18]	EXACT-results	[2]
displacement->infant_birth_weight	0.9541 [0.912, 0.9981]	EXACT	[46]
import_shock->non_displaced_wage_spillover	-0.822 [-1.3042, -0.3398]	EXACT	[6]
family_income_shock->child_achievement_sd	0.061 [0.0157, 0.1063]	EXACT	[27]
family_income_shock->life_expectancy_years	0.1333 [0.1167, 0.15]	EXACT	[22]
displacement->college_enrollment	0.9894 [0.9848, 0.9939]	EXACT	[37]
displacement->parental_income_shortrun	0.864 [0.8568, 0.8712]	EXACT	[37]
unemployment_rate->property_crime	5.018 [2.795, 7.241]	EXACT	[55]
eitc_exposure->adult_earnings_early	564 [83.996, 1,044.004]	EXACT	[7]
male_earnings->marital_fertility	1.24 [0.397, 2.083]	EXACT	[42]
import_shock->gop_win_probability	24.08 [0.423, 47.737]	EXACT	[5]
male_job_loss_rate->child_maltreatment	6 [3.65, 8.35]	EXACT	[47]
displacement->grade_retention_hazard	1.1473 [1.0119, 1.2827]	EXACT	[63]
eviction_order->emergency_shelter_use	4.778 [1.076, 8.48]	EXACT	[25]
eviction_order->eviction_earnings_response	0.8574 [0.7444, 0.9705]	EXACT	[25]
foreclosure_order->house_price_gap	0.99 [0.9875, 0.9925]	EXACT-results	[17]
household_ipv->child_internalizing_sd	0.48 [0.39, 0.57]	EXACT	[31]
household_ipv->child_externalizing_sd	0.47 [0.38, 0.56]	EXACT	[31]
local_unemp_shock->household_ipv	1.58 [1.49, 1.58]	EXACT-results	[59]
household_hardship->household_ipv	2.14 [1.61, 2.68]	EXACT-results	[59]
couple_unemployment->household_ipv	1.3 [1.04, 1.63]	EXACT-results	[59]
unemployment_status->mental_health_sd	0.51 [0.47, 0.54]	EXACT	[54]
import_shock->radical_right_vote_share	13.2 [3.2, 23.2]	EXACT	[24]
neighborhood_exposure->child_outcomes_modifier	0.037 [0.031, 0.043]	EXACT	[20]
unconditional_income->child_education_years	1.127 [0.247, 2.007]	EXACT	[3]
unconditional_income->youth_any_crime	-0.224 [-0.377, -0.071]	EXACT	[3]
unconditional_income->youth_minor_crime_ever	-0.179 [-0.353, -0.005]	EXACT	[3]
youth_crime_conviction_share->youth_crime_convicted	0.061 [0, 0.123]	EXACT	[28]
youth_violent_crime_conviction_share->youth_crime_convicted	0.366 [0.143, 0.589]	EXACT	[28]
displacement->circulatory_mortality_hazard	1.54 [1.29, 1.83]	EXACT	[15]

Link	Point [band]	Tier	Sources
displacement->alcohol_mortality_hazard	1.62 [1.09, 2.41]	EXACT	[15]
displacement->external_cause_mortality_hazard	1.53 [1.15, 2.04]	EXACT	[15]
displacement_event->spouse_mental_health_sd	-0.194 [-0.327, -0.061]	EXACT	[48]
displacement_event->child_achievement_sd	-0.0206 [-0.0402, -0.001]	EXACT	[9]
displacement_event->exam_noncompletion_hazard	1.0688 [1.0014, 1.1362]	EXACT	[9]
displacement_event->hs_completion	-0.037 [-0.0801, 0.0061]	EXACT	[13]
displacement_event->college_attendance	-0.063 [-0.1081, -0.0179]	EXACT	[13]
displacement_event->college_completion	-0.036 [-0.0674, -0.0046]	EXACT	[13]
displacement_event->adult_depression_cesd	0.025 [0.0034, 0.0466]	EXACT	[13]
displacement_event->spouse_participation_elasticity	-0.04 [-0.07, -0.03]	EXACT	[35]
displacement_event->annual_birth_response	-0.005 [-0.0089, -0.0011]	EXACT	[38]
displacement_event->child_depression_anxiety	0.008 [0.0002, 0.0158]	EXACT	[58]
child_achievement_sd->grandchild_achievement_sd	0.38 [0.38, 0.42]	EXACT-results	[12]
child_education_years->grandchild_education_years	0.296 [0.255, 0.337]	EXACT-results	[45]
infant_birth_weight->child_earnings	0.1 [0.0216, 0.1784]	EXACT	[11]
infant_birth_weight->adult_type2_diabetes_hazard	-0.9637 [-1.3834, -0.5401]	EXACT	[43]
infant_birth_weight->adult_cardiovascular_disease_hazard	-0.6994 [-0.8173, -0.585]	EXACT	[43]
divorce_hazard->child_divorce_hazard	1.88 [1.57, 2.24]	EXACT-results	[50]
married_cohort->parental_dissolution_share	0.069 [0.069, 0.072]	EXACT-results	[56]
immigrant_influx->receiving_violent_crime_rate	-0.12 [-0.18, -0.06]	EXACT	[44]
immigrant_influx->receiving_total_crime_rate	0.105 [-0.2616, 0.4716]	EXACT	[8]

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